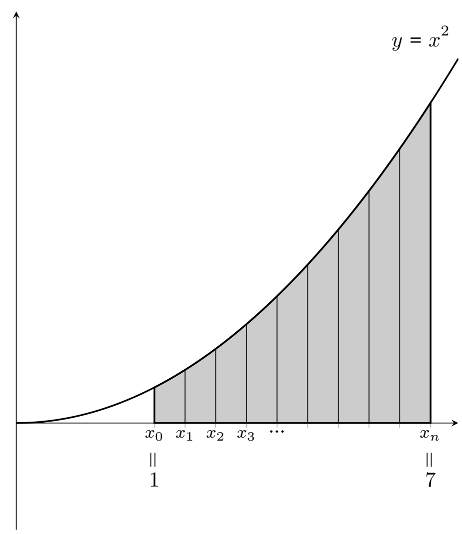
Class 28

Definition and Properties of the Definite Integral

Today's Goals

- You should be able to write a Riemann sum that approximates a definite integral.
- You should be able to state the definition of the definite integral as a limit of Riemann sums and recognize how this definition could be applied to compute a definite integral.
- You should be able to apply the basic properties of the definite integral to simplify expressions involving definite integrals.

1. Hyacinth is interested in finding the area under $y = x^2$ between $x = 1$ and $x = 7$. She decides that she'll first estimate this area using a right-hand sum with n rectangles. So, she slices the area into n slices of equal width Δx. She labels the endpoints of the slices $x_0 = 1, \ldots, x_n = 7$ as shown in the picture below.



   (a) Express Δx in terms of n.

   (b) Express x_1, x_2, and x_{12} in terms of Δx.

   (c) Generalize: express x_k in terms of k and Δx; your formula should work for $k = 0, \ldots, n$.

   (d) Remember that Hyacinth has decided to approximate the area using a right-hand sum. What is the area of the first approximating rectangle? What is the area of the second approximating rectangle? (Express your answers in terms of $x_0, x_1, x_2, \ldots, x_n$ and Δx.)

   (e) Write out R_n, the right-hand sum with n rectangles that approximates the entire area from $x = 1$ to $x = 7$. Your sum should still be in terms of $x_0, x_1, \ldots, x_n$ and Δx.

   (f) Write a limit that gives the *exact* area Hyacinth is looking for.

   (g) Write out L_n in terms of $x_0, x_1, \ldots, x_n$ and Δx.

The Definition of the Definite Integral

The definite integral of $f(x)$ from $x = a$ to $x = b$, denoted $\int_a^b f(x)\,dx$, is defined to be

$$\lim_{n\to\infty} \underbrace{\sum_{k=1}^{n} f(x_k)\Delta x}_{R_n} = \lim_{n\to\infty} \underbrace{\sum_{k=0}^{n-1} f(x_k)\Delta x}_{L_n}, \text{ where } \Delta x = \frac{b-a}{n} \text{ and } x_k = a + k\Delta x$$

provided these limits exist. (If f is continuous on $[a, b]$, then the limits always exist.)

Note on notation: The variable x in $\int_a^b f(x)\,dx$ is a "dummy variable": $\int_a^b f(x)\,dx$, $\int_a^b f(t)\,dt$, and $\int_a^b f(\clubsuit)\,d\clubsuit$ all mean the same thing.

2. Bob is looking at some function $f(x)$ and estimating an area under $y = f(x)$ using a right-hand sum. The sum he's written is

$$R_n = \sum_{k=1}^{n} (\cos x_k)\Delta x \text{ where } \Delta x = \frac{1}{2n} \text{ and } x_k = -\frac{1}{3} + k\Delta x.$$

What area is Bob estimating?

3. Let $f(x) = \cos x$. Suppose we want to approximate $\int_0^{1.5} f(x)\,dx$ using various left-hand and right-hand sums.

(a) Put the following expressions in ascending order.

$$\int_0^{1.5} f(x)\,dx \qquad L_4 \qquad R_4 \qquad L_{20} \qquad R_{20} \qquad L_{100} \qquad R_{100}$$

(b) Find $|R_4 - L_4|$. How can you visualize this? (What's the least work you can do to find $|R_4 - L_4|$? Do you need to calculate R_4 and L_4?)

(c) Find $|R_{100} - L_{100}|$.

(d) How large do we need n to be ensure $|R_n - L_n| < 0.05$? Why might we care to have $|R_n - L_n| < 0.05$?

4. (a) How do $\int_1^4 f(x)\,dx$ and $\int_4^1 f(x)\,dx$ relate? Explain.

(b) Can you simplify $\int_1^3 f(x)\,dx + \int_3^4 f(x)\,dx$? How about $\int_1^3 f(x)\,dx + \int_3^2 f(x)\,dx$?

(c) How does $\int_a^b 3f(x)\,dx$ relate to $\int_a^b f(x)\,dx$? Why?

(d) How does $\int_a^b [f(x)+g(x)]\,dx$ relate to $\int_a^b f(x)\,dx$ and $\int_a^b g(x)\,dx$? Why?

Properties of the Definite Integral

- $\int_b^a f(x)\,dx =$ ________________
- For a constant k, $\int_a^b k \cdot f(x)\,dx =$ ________________
- $\int_a^b f(x)\,dx + \int_b^c f(x)\,dx =$ ________________
- $\int_a^b [f(x)+g(x)]\,dx =$ ________________________

5. Suppose you know that $\int_2^5 f(t)\,dt = 7$. Do you have enough information to evaluate each of the following integrals? If so, evaluate; if not, explain why not.

   (a) $\int_2^5 [f(t)+4]\,dt$

   (b) $\int_2^5 f(t+4)\,dt$

   (c) $\int_5^2 4f(t)\,dt$

   (d) $\int_2^4 [f(t)+t]\,dt + \int_4^5 f(t)\,dt$